Practice Problem 12.1

In return for payments of \$5,000 at the end of 3 years and \$4,000 at the end of 9 years, an investor agrees to pay \$1,500 immediately and to make an additional payment at the end of 2 years. Find the amount of the additional payment if $i^{(4)} = 0.08$.

Solution

The nominal annual interest rate convertible quarterly is 8%, so the quarterly rate is $0.08/4 = 0.02$. The effective annual rate is

$$\begin{aligned} i &= \left(1 + \frac{0.08}{4}\right)^4 - 1 \\ &= 1.02^4 - 1 \\ &= 0.08243216. \end{aligned}$$

Let K be the additional payment due at the end of 2 years. Equating the present values at time 0 gives

$$\begin{aligned} 1{,}500 + \frac{K}{(1.02)^8} &= \frac{5{,}000}{(1.02)^{12}} + \frac{4{,}000}{(1.02)^{36}}, \\ K &= \left(\frac{5{,}000}{(1.02)^{12}} + \frac{4{,}000}{(1.02)^{36}} - 1{,}500\right)(1.02)^8 \\ &\approx 5{,}159.23627. \end{aligned}$$

Therefore, the additional payment is approximately

$$\boxed{\$5{,}159.24}.$$